

Zihao Zhang

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PROFILE

I am a master's student in the School of Computer Science at Fudan University, advised by [Prof. Yu-Gang Jiang](#) and [Prof. Zuxuan Wu](#).

My research explores generative visual modeling for **embodied intelligence**, with the goal of building **controllable and efficient visual foundations** that help embodied agents understand and predict how scenes evolve over time.

My work has been published at leading computer vision and multimedia venues, including **CVPR**, **ICCV**, **ACM MM** and **SIGGRAPH Asia**.

RESEARCH INTERESTS

- **Generative world models** and controllable video generation for embodied intelligence
- **Spatial reasoning**, camera planning, and motion understanding
- **Efficient diffusion models** for high-resolution visual generation

EDUCATION

Sep 2023 — Jun 2026 **Fudan University (QS 26)**

Master's Degree · Shanghai, China

School of Computer Science · GPA: 3.5 · Advisors: Prof. Yu-Gang Jiang & Prof. Zuxuan Wu

Sep 2018 — Jun 2022 **Wuhan University (QS 165)**

Bachelor's Degree · Wuhan, China

School of Chemistry and Molecular Sciences

AWARDS & HONORS

2025 **First-Class Academic Scholarship** · Fudan University

2020 & 2021 **First-Class Scholarship and Grant** · Wuhan University

2025 **EDEN Patent Application** · Huawei Noah's Ark Lab · Large-motion video frame interpolation

RESEARCH EXPERIENCE

Apr 2025 — Mar 2026 Research Intern · **Bilibili Inc.** Shanghai, China
SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation — Co-first author, ACM MM 2026

- Led the one-step pixel-space diffusion framework and designed a progressive multi-stage DiT, Noise-Update-Only Attention, and Drift-aware Timestep Sampling to learn motion, structure, and appearance from coarse to fine.
- Achieved state-of-the-art visual-dynamics prediction: 8.8% lower LPIPS on SNU-FILM, 63.3% faster inference, 10.6% less memory, and up to 51.5% lower LPIPS on 4K benchmarks.

CT-1: Vision-Language-Camera Models for Camera-Controllable Video Generation — Co-first author, ACM MM 2026

- Co-developed a Vision-Language-Camera model that grounds visual observations and language instructions in SE(3) camera trajectories, enabling spatially aware and temporally stable viewpoint planning.
- Contributed to the CT-200K curation pipeline (>47M frames) and controllable video-diffusion system, improving camera-control success by 25.7% and advancing spatially grounded generative world modeling.

Jul 2024 — Mar 2025 Research Intern · **Huawei Noah's Ark Lab** Advised by [Hang Xu](#) · Shanghai, China
EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation — First author, CVPR 2025

- Led EDEN's architecture and training, developing a Transformer Tokenizer, dual-stream temporal attention, and frame-difference conditioning for large, nonlinear motion.
- Reduced LPIPS by nearly 10% on DAVIS and SNU-FILM and by about 8% on DAIN-HD, improving high-fidelity temporal state prediction under fast scene changes.

MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion — Third author, ICCV 2025

- Contributed lightweight pose/reference controllers and two-branch score guidance to transfer target body motion while retaining subject identity, background details, and camera dynamics.
- Reduced GPU memory by approximately 80% versus MotionEditor while improving videos with large camera motion and complex backgrounds, enabling efficient motion-conditioned human-scene synthesis.

Publications

- 01 **EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation**
Zihao Zhang, Haoran Chen, Haoyu Zhao, Guansong Lu, Yanwei Fu, Hang Xu, Zuxuan Wu
CVPR 2025

- 02 **SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation**
Zihao Zhang*, Haoyu Zhao*, Siqian Yang, Yidi Wu, Yudong Jiang, Zuxuan Wu
ACM MM 2026

- 03 **CT-1: Vision-Language-Camera Models Transfer Spatial Reasoning Knowledge to Camera-Controllable Video Generation**
Haoyu Zhao*, Zihao Zhang*, Jiaxi Gu, Haoran Chen, Qingping Zheng, Pin Tang, Yeying Jin, Yuang Zhang, Junqi Cheng, Zenghui Lu, Peng Shu, Zuxuan Wu, Yu-Gang Jiang
ACM MM 2026

- 04 **MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion**
Shuyuan Tu, Qi Dai, Zihao Zhang, Sicheng Xie, Zhi-Qi Cheng, Chong Luo, Xintong Han, Zuxuan Wu, Yu-Gang Jiang
ICCV 2025

- 05 **VIDiff: Translating Videos via Multi-Modal Instructions with Diffusion Models**
Zhen Xing, Qi Dai, Zihao Zhang, Hui Zhang, Han Hu, Zuxuan Wu, Yu-Gang Jiang
arXiv:2311.18837, 2023

- 06 **AniME: Adaptive Multi-Agent Planning for Long Animation Generation**
Lisai Zhang, Baohan Xu, Siqian Yang, Mingyu Yin, Jing Liu, Chao Xu, Siqi Wang, Yidi Wu, Yuxin Hong, Zihao Zhang, Yanzhang Liang, Yudong Jiang
SIGGRAPH Asia 2025 Posters

- 07 **EVA: Enhancing Anime Video Generation via Reinforcement Learning**
Yidi Wu, Bingwen Zhu, Zihao Zhang, Siqian Yang, Lisai Zhang, Baohan Xu, Mingyu Yin, Yanzhang Liang, Yudong Jiang
ICASSP 2026